

North South University

Department of Electrical and Computer Engineering (ECE)



Course Name: Internet and Web Technology

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Project Proposal

Project Title: AI-Enabled Multivendor Learning Management System (LMS)

Group Number: 1

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1 User Stories

Objective: User stories describe the system from the perspective of different users to help identify all functionalities that the application should provide.

1.1 Guest Users

- As a guest, I want to browse available courses so that I can see what is offered before signing up.
- As a guest, I want to search for courses by category so that I can find topics of interest.
- As a guest, I want to view instructor profiles so that I can check their credentials.
- As a guest, I want to register as a student or instructor so that I can access platform features.
- As a guest, I want to view course ratings and reviews so that I can choose the best course.
- As a guest, I want to filter courses by price and category so that I can quickly find suitable courses.
- As a guest, I want to contact instructors so that I can ask questions before enrolling.
- As a guest, I want to view featured and trending courses so that I can discover popular content.

1.2 Student (Customer)

- As a student, I want to log in securely so that my learning progress is saved.
- As a student, I want to enroll in a course so that I can access its learning materials.
- As a student, I want to pay for premium courses using an integrated payment gateway so that my enrollment is instantly confirmed.
- As a student, I want to watch video lectures so that I can learn at my own pace.
- As a student, I want to receive AI-based course recommendations so that I can discover relevant content based on my interests.
- As a student, I want to take quizzes so that I can test my knowledge.
- As a student, I want to ask questions to an AI chatbot so that I can get immediate clarifications on course topics.
- As a student, I want to leave a review and rating for a course so that other students can make informed decisions.
- As a student, I want to view my dashboard so that I can track my overall course completion progress.

- As a student, I want to download a certificate upon course completion so that I can share my achievement.
- As a student, I want to bookmark courses so that I can enroll in them later.
- As a student, I want to receive notifications about course updates so that I stay informed.
- As a student, I want to reset my password so that I can recover my account.
- As a student, I want to track my quiz scores so that I can evaluate my performance.
- As a student, I want to download course materials so that I can study offline.
- As a student, I want to participate in discussion forums so that I can interact with instructors and peers.
- As a student, I want to view my purchase history so that I can manage my enrolled courses.
- As a student, I want to edit my profile information so that I can keep my account updated.

1.3 Instructor (Vendor)

- As an instructor, I want to create a new course listing so that I can offer my knowledge to students.
- As an instructor, I want to upload video lessons and PDF resources so that students have comprehensive study materials.
- As an instructor, I want to use AI to automatically generate quiz questions based on my course description so that I save time on course creation.
- As an instructor, I want to set the price of my course so that I can monetize my content.
- As an instructor, I want to view my earnings dashboard so that I can track my revenue and enrollments.
- As an instructor, I want to reply to student reviews so that I can engage with my audience.
- As an instructor, I want to submit my course for admin approval so that it can be published on the marketplace.
- As an instructor, I want to edit my course content so that I can keep my courses up to date.
- As an instructor, I want to create assignments and quizzes so that I can assess students.
- As an instructor, I want to view analytics about student engagement so that I can improve my courses.

- As an instructor, I want to issue certificates so that students receive proof of completion.
- As an instructor, I want to receive notifications when students enroll so that I can monitor my courses.
- As an instructor, I want to save courses as drafts so that I can publish them later.
- As an instructor, I want to manage discount offers so that I can attract more students.

1.4 Admin

- As an admin, I want to review pending courses so that I can approve or reject them based on quality guidelines.
- As an admin, I want to manage user accounts (students and instructors) so that I can ban or suspend users who violate terms of service.
- As an admin, I want to view platform-wide financial reports so that I can track overall revenue and commission payouts.
- As an admin, I want to manage course categories so that the marketplace remains organized.
- As an admin, I want to generate reports so that I can analyze platform performance.
- As an admin, I want to manage featured courses so that high-quality courses receive visibility.
- As an admin, I want to send announcements to all users so that important information reaches everyone.
- As an admin, I want to monitor user activities so that suspicious behavior can be detected.
- As an admin, I want to manage payment transactions so that refunds and disputes can be handled.
- As an admin, I want to configure platform settings so that the system operates correctly.

2 Software Requirements Specification (SRS)

Objective: The SRS formally describes the entire software system, including its requirements and expected behavior.

2.1 1. Introduction

Project Title: AI-Enabled Multivendor Learning Management System (LMS)

Team Members: Md Abir Hossain, Sabrina Sabnam Farhana, Laskar Fahad Ahmed Rumi

Problem Statement: Traditional e-learning platforms often lack personalized learning paths and automated tools for instructors. Additionally, creating a marketplace where multiple independent instructors can easily host and sell their courses requires a robust, scalable system with strict quality control.

Objectives: To build a scalable, multivendor LMS where instructors can seamlessly create and monetize courses, while students benefit from AI-driven recommendations, AI tutoring, and a seamless learning experience.

Scope of the project: The system will cover user authentication, course creation and approval workflows, payment processing, video streaming, AI-powered quiz generation, and progress tracking.

2.2 2. System Overview

Purpose: A platform bridging independent educators and learners through an intelligent digital marketplace.

Target Users: Independent educators (Instructors), global learners (Students), and platform operators (Admins).

Expected Outcome: A fully functional web application capable of handling course hosting, multi-user role management, financial transactions, and AI integrations.

2.3 3. Functional Requirements

- **FR-01 User Registration & Login**

Description: Allows users to sign up and authenticate as Student or Instructor.

Inputs: Name, Email, Password, Role.

Outputs: Authentication token, Dashboard access.

Preconditions: User must not have an existing active account with the same email.

Postconditions: User session is established.

- **FR-02 Course Creation Workflow**

Description: Instructors can draft courses, upload media, and submit for review.

Inputs: Course Title, Description, Price, Video files, Documents.

Outputs: Pending Course Object in database.

Preconditions: User must be logged in as an Instructor.

Postconditions: Course status set to 'Pending Review'.

- **FR-03 AI Quiz Generation**

Description: System uses AI to read course descriptions/transcripts to generate multiple-choice questions.

Inputs: Course text/context.

Outputs: JSON array of questions, options, and correct answers.

Preconditions: Instructor requests quiz generation during course creation.

Postconditions: Quiz is attached to the course module.

- **FR-04 Admin Verification Workflow**

Description: Admins can view pending courses and approve/reject them.

Inputs: Admin decision (Approve/Reject), Feedback text.

Outputs: Updated course status, Email notification to instructor.

Preconditions: Course must be in 'Pending Review' state.

Postconditions: Course becomes publicly visible (if approved).

- **FR-05 Course Enrollment & Payment**

Description: Students can securely purchase a course.

Inputs: Payment details.

Outputs: Transaction receipt, updated enrollment list.

Preconditions: User logged in as Student, valid payment method.

Postconditions: Student gains lifetime access to course materials.

2.4 4. Non-Functional Requirements

- **Performance:** The platform should load pages within 2 seconds under normal load.
- **Security:** Passwords must be hashed using bcrypt; API endpoints must be protected via JWT.
- **Availability:** System should target 99.9% uptime.
- **Scalability:** Database and backend should handle a growing number of simultaneous video streams and users.
- **Usability:** Intuitive UI/UX for non-technical instructors and students.
- **Responsiveness:** Fully responsive design for Mobile, Tablet, and Desktop.

2.5 5. Technologies

- **Frontend:** React.js, Tailwind CSS
- **Backend:** Node.js, Express.js
- **Database:** MySQL (Relational structure for financial and enrollment integrity)
- **Authentication:** JWT (JSON Web Tokens)
- **AI Integration:** OpenAI API (for Chatbot and Quiz Generation)
- **Deployment:** Frontend on Vercel, Backend on Render/AWS.

2.6 6. Assumptions and Constraints

- Continuous internet connection required.
- Modern browser support (Chrome, Firefox, Safari, Edge).
- Video hosting may rely on a third-party service (e.g., AWS S3 or Vimeo API) to reduce server load.

3 Wireframes

Objective: Illustrate how the application will look focusing on layout and functionality.

3.1 Landing Page and Authentication

Navigation bar (Logo, Categories, Search, Login/Register). Hero section with a call to action. Horizontal scrolling cards for 'Top Rated Courses' and 'AI Recommendations'. Includes gateway windows managing system access protocols.

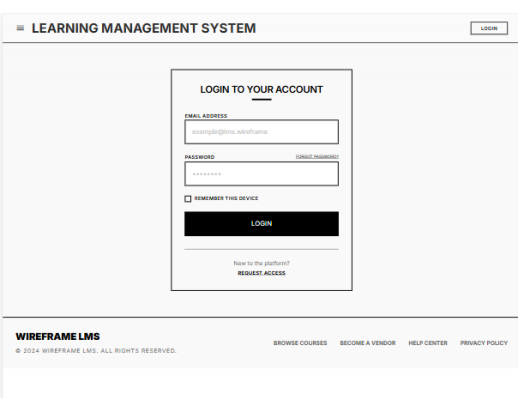


Figure 1: Login Interface

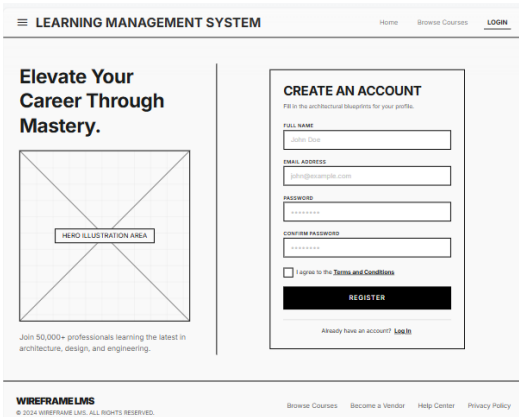


Figure 2: Account Creation Portal

3.2 Student Dashboard

Sidebar with (My Courses, Wishlist, Profile, Settings). Main area displaying progress bars for currently enrolled courses and recent AI-suggested content.

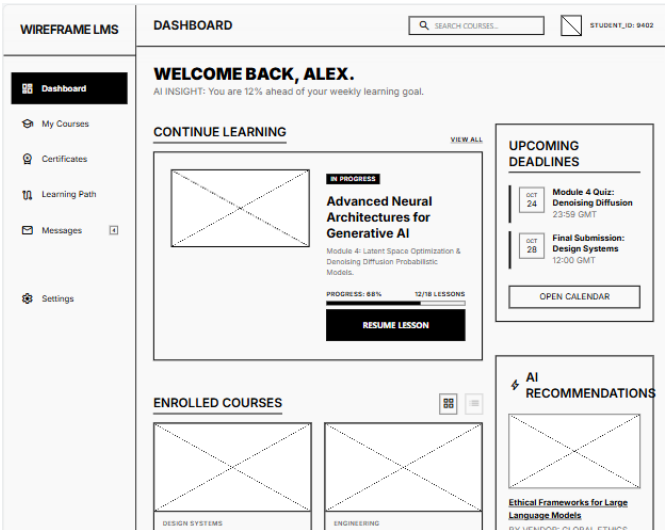


Figure 3: Student Progress Tracking Dashboard

3.3 Instructor Dashboard

Sidebar with (My Courses, Create Course, Earnings, Reviews). Top KPI cards (Total Students, Total Revenue, Active Courses). A data table displaying course statuses (Approved, Pending, Draft).

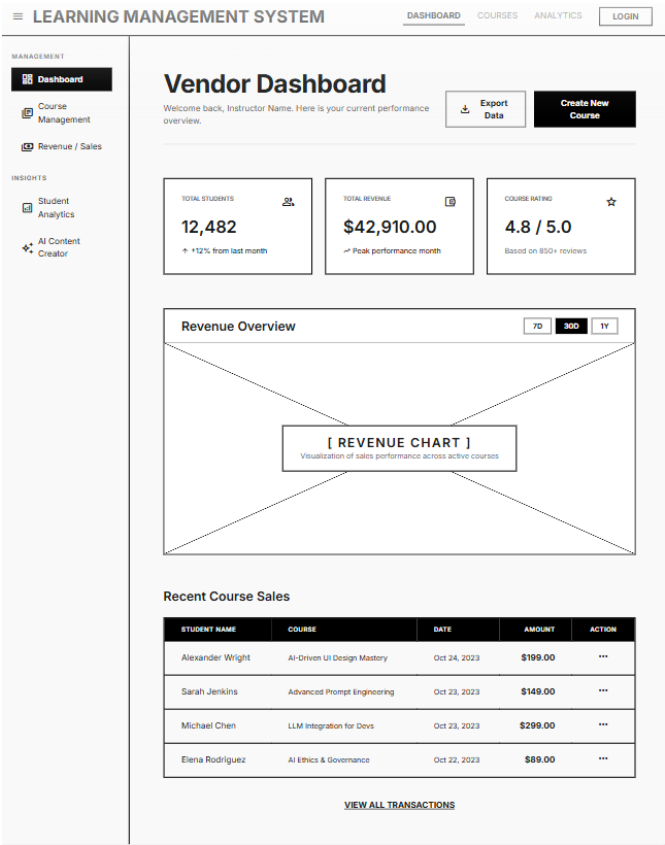


Figure 4: Instructor Operational Analytics Interface

3.4 Course Details Page

Large video player placeholder at the top. Course title, instructor bio, price, and “Enroll Now” button on the right sticky sidebar. Tabbed content below for Overview, Curriculum, Reviews, and AI QA Bot. The second layout frame captures visual configuration context shifts upon natural scrolling.

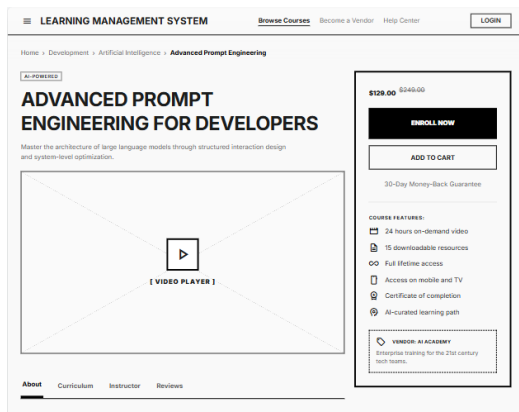


Figure 5: Course View (Top Panel)

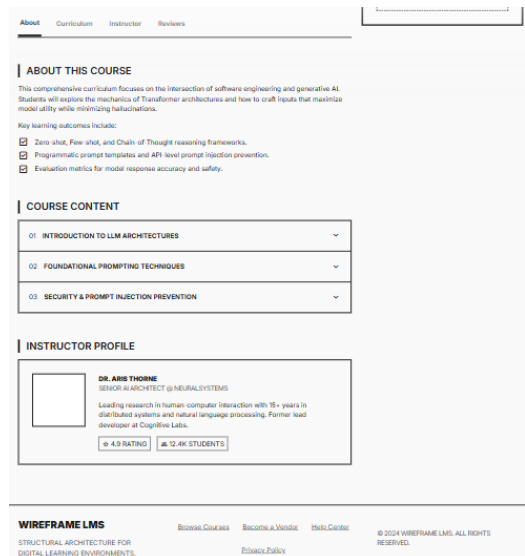


Figure 6: Course View (Scrolled Context)

3.5 Checkout Screen

Secure payment terminal wireframe supporting dynamic totals, discount application entries, and transaction configuration tools.

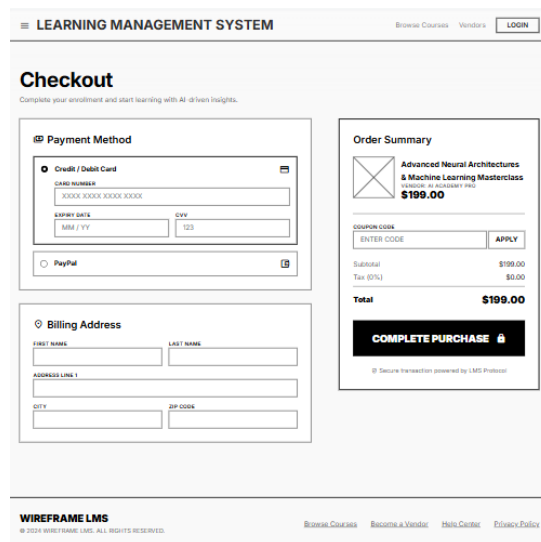


Figure 7: Payment Operations Interface

3.6 Admin Dashboard

Global sidebar (Users, Courses, Revenue, Settings). Main table view for 'Pending Course Approvals' with quick action buttons (Review, Approve, Reject).

4 Database Schema

Objective: Design the database demonstrating how data will be stored securely and relationally using MySQL.

4.1 Entity List & Attributes

- **Users:** `user_id` (PK), `name`, `email`, `password_hash`, `role` (Enum: Admin, Instructor, Student), `created_at`.
- **Courses:** `course_id` (PK), `instructor_id` (FK), `title`, `description`, `price`, `status` (Enum: Draft, Pending, Approved), `thumbnail_url`, `created_at`.
- **Modules (Curriculum):** `module_id` (PK), `course_id` (FK), `title`, `video_url`, `duration`, `sequence_order`.
- **Enrollments:** `enrollment_id` (PK), `student_id` (FK), `course_id` (FK), `progress_percentage`, `enrolled_at`.
- **Payments:** `payment_id` (PK), `enrollment_id` (FK), `amount`, `payment_status`, `transaction_date`.
- **Reviews:** `review_id` (PK), `course_id` (FK), `student_id` (FK), `rating` (1-5), `comment`, `created_at`.

4.2 Relationships

- **Users to Courses (Instructor):** One-to-Many (One Instructor can create many Courses).
- **Courses to Modules:** One-to-Many (One Course contains multiple Modules).
- **Users to Enrollments (Student):** One-to-Many (One Student can have multiple Enrollments).
- **Courses to Enrollments:** One-to-Many (One Course can have multiple Enrollments).
- **Enrollments to Payments:** One-to-One (One Enrollment corresponds to one Payment record).
- **Courses to Reviews:** One-to-Many (One Course can have many Reviews).

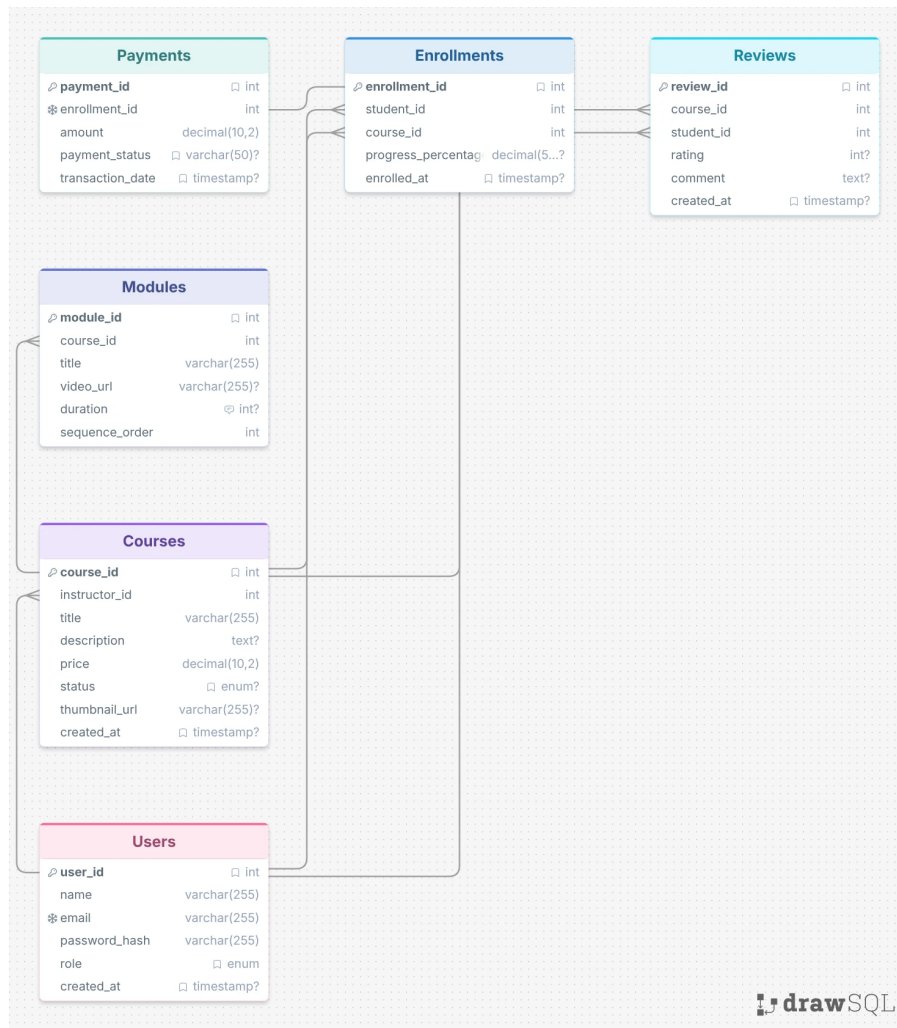


Figure 8: Entity Relationship (ER) Diagram for the LMS

5 Project Quotation

5.1 Part A: Quotation Letter

Group-1

Department of ECE, North South University
Bashundhara, Dhaka-1229
Email: abir.hossain06@nortsouth.edu

Date: August 20, 2026

Client Name: Mr. Kazi Tanvir Akter

Company Name: EduTech Innovations Ltd.

Address: Gulshan Avenue, Dhaka

Subject: Quotation for the Development of AI-Enabled Multivendor LMS Web Application

Dear Mr. Tanvir,

We are pleased to submit our quotation for the development of the proposed AI-Enabled Multivendor Learning Management System. Based on the project requirements, we have prepared a comprehensive solution that meets your business needs for a modern, scalable educational marketplace.

The proposed system includes role-based dashboards (Admin, Instructor, Student), secure payment gateways, a course verification workflow, and advanced AI features such as automated quiz generation and personalized course recommendations. Our technology stack utilizing React.js, Node.js, and MySQL ensures high performance and security.

The estimated cost of the project is **BDT 450,000**, with an estimated development duration of **10 weeks**. A detailed cost breakdown is attached with this quotation.

Thank you for considering our proposal. We appreciate the opportunity to submit this quotation and look forward to working with you. Please feel free to contact us if you require any clarification regarding the proposed solution.

Sincerely,

Md Abir Hossain

Team Leader, Group-1

August 20, 2026

5.2 Part B: Cost Estimation Summary

Project Information

Project Name: AI-Enabled Multivendor LMS

Client Name: Mr. John Doe (EduTech Innovations Ltd.)

Team Name: Group-1 Web Solutions

Development Timeline:

The project will be executed over a 10-week timeline, divided into the following phases:

- Requirement Analysis & UI/UX Design (Weeks 1-2)
- Database Design & Authentication (Week 3)
- Backend API Development (Weeks 4-5)
- Frontend Development & Integration (Weeks 6-8)
- Testing, QA & AI Integrations (Week 9)
- Deployment & Documentation (Week 10)

Component	Estimated Cost (BDT)
Requirement Analysis	25,000
UI/UX Design	45,000
Frontend Development	110,000
Backend Development	120,000
Database Design	40,000
Authentication & Security	30,000
Testing & Quality Assurance	40,000
Deployment	20,000
Documentation	20,000
Maintenance (Optional)	TBD
Total Estimated Cost	450,000

Cost Summary

- **Total Estimated Cost:** BDT 450,000
- **Estimated Development Time:** 10 Weeks
- **Team Size:** 3 Developers
- **Currency:** BDT (Bangladeshi Taka)

Assumptions

- Three-member development team working part-time over 10 weeks.
- Excludes costs for premium third-party APIs (e.g., OpenAI API usage costs) and cloud hosting subscriptions (AWS/Vercel) post-deployment, which will be billed directly to the client.
- Excludes post-deployment maintenance unless a separate maintenance contract is signed.